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Document representation and feature combination for deceptive spam review detection

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ABSTRACT

Deceptive spam reviews of products or service are harmful for customers in decision making. Existing approaches to detect deceptive spam reviews are concerned in feature designing. Hand-crafted features can show some linguistic phenomena, however can hardly reveal the latent semantic meaning of the review. We present a neural network based model to learn the representation of reviews. The model makes a hard attention through the composition from sentence representation into document representation. Specifically, we compute the importance weights of each sentence and incorporate them into the composition process of document representation. In the mixed-domain detection experiment, the results verify the effectiveness of our model by comparing with other neural network based methods. As the feature selection is very important in this direction, we make a feature combination to enhance the performance. Then we get 86.1% F1 value which outperform the state-of-the-art method. In the cross-domain detection experiment, our method has better robustness.

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1. Introduction

Deceptive opinion spam detection is an urgent and meaningful task in the field of natural language processing. By continuous growth of the user-generated reviews, the appearance of deceptive opinion spam arouses people's attention [1–4]. Deceptive opinion spam is the review with fictitious opinions which are deliberately written to sound authentic [5]. For commercial motive, some businesses hire people to write undeserving positive reviews to promote the objects or unjust negative reviews to damage the reputations of the objects [6]. It is very difficult for people to distinguish deceptive spam. In the test of Ott et al. [5], the average accuracy of three human judges is only 57.33%. Hence, the research in detecting deceptive opinion spam is necessary and meaningful.

The reviews are commonly short documents. The objective of the task is to distinguish whether the document whether is a spam or a truth. The task can be transformed into a 2-category classification problem. The majority of existing approaches follows Ott et al. [5] and utlizes machine learning algorithms to build the classifiers. Under this direction, most studies focus on designing effective features to enhance the classification performance. Feature engineering is important, however, we can hardly learn the inherent law of data from a semantic perspective. In view of the good performance of neural network based models in the natural language processing tasks currently, the document-level representation can be learnt by neural network based models, and be used as features of the review.

In this work, we try to make a comparison and analysis between representation learning algorithms and conventional features while solving the problem. We present a novel method which is sentence weighted neural network (SWNN) model to learn the document-level representation of the review and detect spam reviews. Learning the representation of the document can capture the global feature and take word order and sentence order into consideration. We also make a feature combination with SWNN, that the features are firstly used jointly in the spam review detection. We verify the effectiveness of SWNN and the feature combination in two types of experiments. One is to verify the capability of domain migration on cross-domain dataset, and another is to verify the capability of domain-independent spam review detection on mix-domain dataset. The experiments run on the public data sets [7]. The domain migration experiment verifies that feature combination with SWNN has the best robustness. The domain-independent experiment verifies that the feature combination with unigram perform better than the feature combination with SWNN. The final result outperforms other strong baseline methods with the highest F1 value of 86.1%.

The major contributions of the work presented in this paper are as following:

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- We present a sentence weighted neural network to learn the representation of document-level reviews. To learn the semantic of the document better, the proposed model takes the importance of different sentences through compositing sentence representation into document representation.
- We use multiple syntax features and make a feature combination to further improve the performance of our method.
- We verify the effectiveness of SWNN and feature combination in domain migration and domain-independent experiments for spam review detection.

It should be noted that the work is the extension of our last work which is about learning document representation for deceptive opinion spam detection [8]. In the last work, we presented the SWNN model to learn document representation and detect deceptive spam reviews. In this work, we make two improvements. First, we introduce some new syntax features and use the feature combination to resolve the problem. Second, we incorporate syntax features with SWNN to jointly detect spam reviews. The experimental result outperforms the original one.

2. Related work

We present a brief review of the related work from two perspectives. One is deceptive opinion spam detection, and the other is neural networks for specific task representation learning.

2.1. Deceptive opinion spam detection

On the Internet, various kinds of spam bring troubles to people. Over the years, many studies focus on spam detection. Web spam has been extensively studied [9–15]. The objective of the web spam is to gain high page rank and attract people to click by fooling search engines. Email spam is another related research, which pushes unsolicited advertisements to users [16,17]. Social media spam is a type of spam information which spreads rumors on the social media [18]. The web spam and mail spam have a common character that they have irrelevant words. Opinion spam is quite different and more crafty which contains opinions of users about products and services. By the explosive growth of user-generated content, the number of opinion spam in the reviews increases continuously. This phenomenon attracts the attention of researchers. Opinion spam is firstly investigated by Liu et al. [6] who also summarize the opinion spam into different types. In terms of different damages to users, we can further conclude the opinion spam into two types which are including deceptive opinion spam and product-irrelevant spam. In the former spam, the spammers give undeserving positive reviews or unjust negative reviews to the object for misleading costumers. The latter spam contains no comments about the object. Obviously, the deceptive opinion spam is more difficult to detect.

The approaches to detect deceptive opinion spam can be divided into unsupervised methods and supervised methods. Liu et al. [19] take a Bayesian approach and formulate opinion spam detection as a clustering problem. There are also many unsupervised methods to research on spammers detection [20–23] and reviewing patterns mining [24]. Due to the lack of gold standard data, most methods take researches on pseudo labeled data. Liu et al. [6] assume duplicate and near duplicate reviews to be deceptive spam. They also apply features of review texts, reviewers and products. Yoo et al. [25] first collect a small amount of deceptive spam and truth reviews and do a linguistic analysis on them. By applying Amazon Mechanical Turk, Ott et al. [5,26,27] gather a gold standard labeled data. A few follow-up spam detecting methods have been presented on the data set. Ott et al. estimate prevalence of deceptive opinion spam in reviews [26], and identify negative spam [27]. Li et al. [28] identify manipulated offerings on review portals. Feng et al. [29] apply context free grammar parse trees to extract syntactic features to improve the performance of the model. Feng and Hirst [30] take the group of reference reviews into account according to the same product. Although there are deceptive opinion spam in the Ott's data sets, it still can not reflect the real condition with the lack of cross-domain data, and the Turkers also lack of professional knowledge. Li et al. [7] create a cross-domain data sets (i.e. hotel, restaurant, and doctor) with part of reviews from domain experts. On this labeled data set, they use n-gram features as well as POS and LIWC features in classification and show that POS perform more robust on cross-domain data.

2.2. Neural networks for representation learning

Representation learning by neural networks based methods have been proven to be effective in the place of task-specific feature engineering [31]. Compared with feature engineering, representation learning does not need much prior knowledge. As a continuous real-valued vector, representation can be incorporated as features in a variety of natural language processing tasks [32–34], such as POS tagging, chunking, named entity recognition [32,35], semantic role labeling, parsing [36], language modeling [37,38], sentiment analysis tasks [39,40] and text classification [41]. Representation learning is to learn continuous representations of text with different grains, like word, phrase, sentence and document.

As for representing a document, the existing deep learning methods consist of two processing stages. Firstly, word embedding should be learnt by massive text corpus. Some work utilizes global context of document and multiple word prototypes [42], or global word-word co-occurrence to improve word embedding [43]. There are also some work for task-specific word-embedding [40]. After obtaining word representation, many studies focus on researching the semantic composition methods. Yessenalina et al. use matrices to model each word and applying iterative matrix multiplication to combine words [44]. Glorot et al. develop Stacked Denoising Autoencoders for domain adaptation [45]. Socher et al. propose Recursive Neural Network (RNN) [46], matrixvector RNN [47] and Recursive Neural Tensor Network (RNTN) [39] to learn the semantic of unfixed-length phrases. Hermann et al. [48] learn the semantic of sentences by Combinatory Categorial Autoencoder method. The method is the combination of Combinatory Categorial Grammar and Recursive Autoencoder. Li et al. [49] use feature weight tuning to control the effect one specific unit makes to the higher-level representation in a Recursive Neural Network. Le et al. [50] learn the representation of paragraph.

3. Methodology

In the section, we present the details of neural network based models to learn document representation for deceptive spam review detection. We develop two convolutional neural network models to learn document representation. In the following sections, we firstly introduce the conventional model and then present the details of our proposed models.

3.1. Basic convolutional neural network

Collobert et al. [32] introduce a neural network approach to learn the representation of a sentence. The architecture is shown in Fig. 1. It is a multilayer neural network which consists of four layers. Given a sentence "*The Chicago Hilton is very great*", the model applies the lookup layer to map these words into corresponding word embeddings which are continuous real-valued vectors. The convolutional layer extracts local features by representing the semantic meaning of the words in the window. The

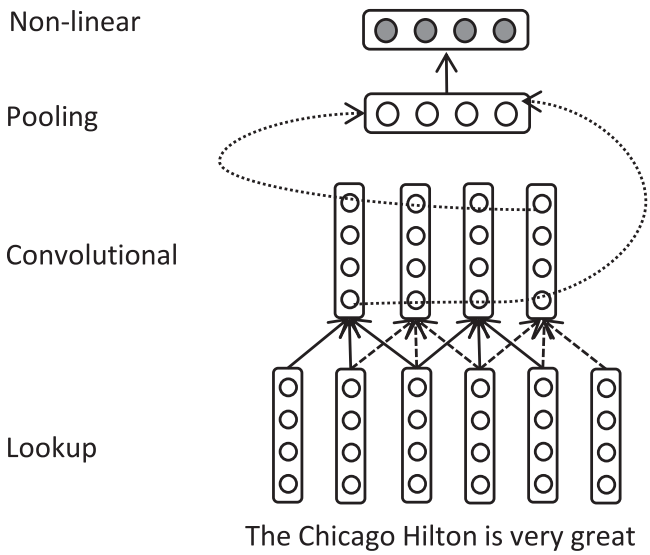

Fig. 1. The traditional neural network for learning sentence representation.

size of the output of convolutional layer depends on the number of words in the sentence fed to the network. Pooling layer obtains a global feature vector by combining the local feature vectors through previous layers. Common operations are doing average or max operations over the corresponding vectors. The average operation captures the influence of all words to the certain task. The max operation captures the most useful local features produced by convolutional layer. The non-linear layer is necessary to extract high level features.

3.2. The document representation learning model

Basic model. We apply the traditional convolutional neural network model to represent sentences. To make a composition for the document, we use average operation to capture all of the sentences features on the pooling layer. This is a basic model, which is modified below to suit the deceptive opinion spam detection task.

SCNN model. As the architecture is shown in Fig. 2, SCNN model consists of two convolutional layers to do the composition. The sentence convolution is to make a composition of each sentence by a fix-length window. The document convolution transforms sentence vectors into a document vector. The output is consisting of the scores of the corresponding category. The training objective is that the gold label has a higher score than the other label through predicting the label of a review. This objective can be optimized by a hinge loss. Compared with softmax function which is with strict constrain, hinge loss is a relaxed constraint that is more suitable for our objective. Hinge loss function is defined as shown in Eq. (1), where t is the gold label of the review r, t^* stands for the another label, and m_δ is the margin in the experiment.

$$Loss(r) = max(0, m_\delta - f(r_t) + f(r_{t^*})) \tag{1}$$

SWNN model. The sentence-weighted neural network model is a modified model of the basic document representation learning model. As a matter of fact, the words in a review play different roles in the semantic representation. Some words must be more important in distinguishing spam from the truth reviews. Hence, each sentence also owns its importance weight according to the words in it. We compute the importance weight of the sentence based on the importance weights of words in the sentence. We apply KL-divergence as the importance weight of the word. The value of KL-divergence stands for the capacity of a feature in dividing documents which is a feature selection approach. In fact, we also try $tf - idf$ as a candidate of weight computing method, however, it does not perform as well as KL-divergence in the experiment. We assume that $U = \{U_1, \ldots, U_i, \ldots, U_n\}$ is the universal set of words in the review, where U_i is the word set of the ith sentence, and W_j stand for the weight of the jth word. The sentence weight is a normalization value like in the following formula.

$$\alpha_i = \frac{\sum_{j \in U_i} W_j}{\sum_{k \in U} W_k} \tag{2}$$

In Fig. 3, the architecture of SWNN model is given. Each sentence of the input document review transforms into the fixed-length vector through convolutional layer. The process of generating sentence weights produce normalized weight α_i corresponding to the ith sentence. Through the pooling layer, the sentence vectors transform into a document vector by a weighted-average operation. More important sentences have more influences when producing the document vector. The vector transforms through tanh layer to extract high level features. The linear layer produces the scores of the categories.

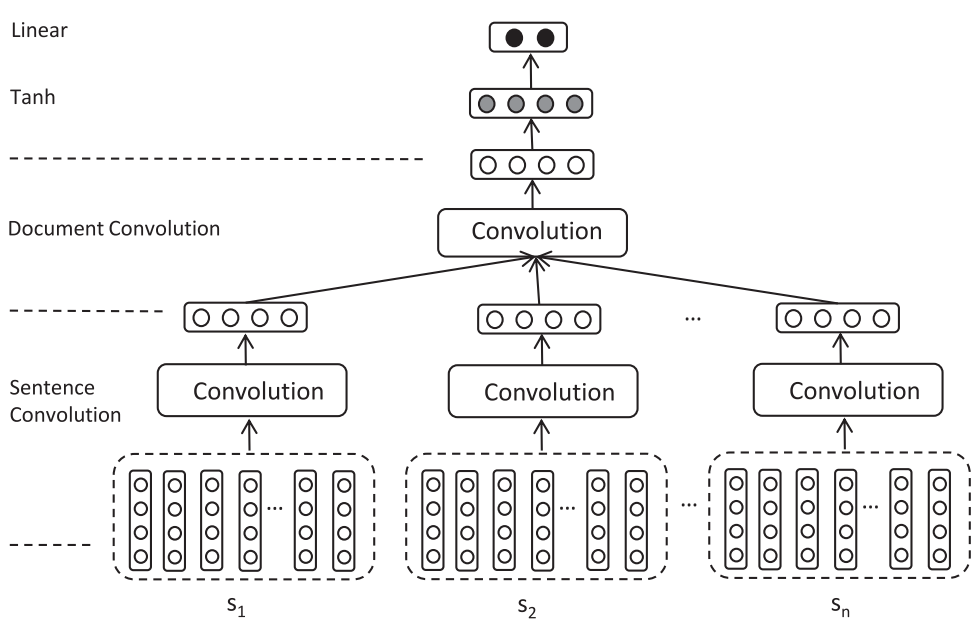

Fig. 2. SCNN model for learning sentence representation.

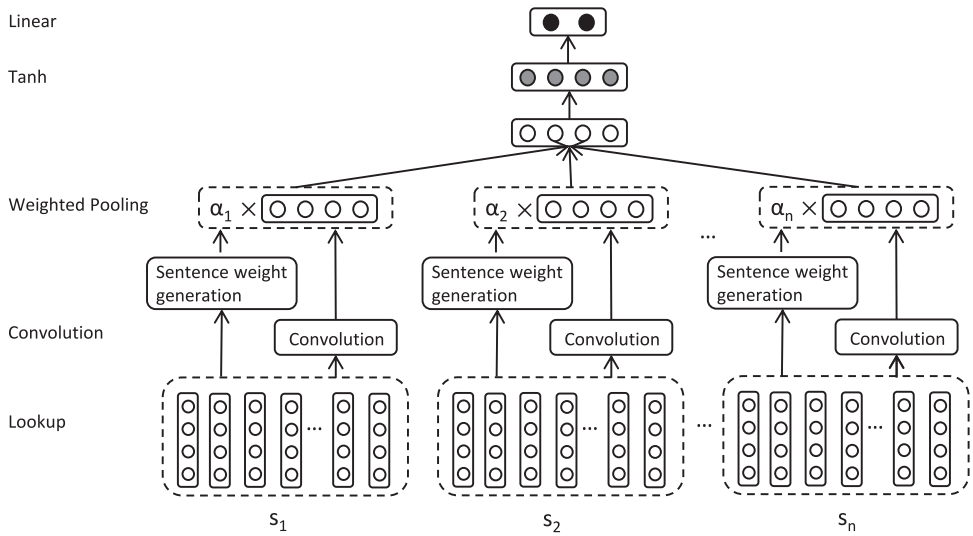

Fig. 3. SWNN model for learning sentence representation.

3.3. Features

We add two types of features to the proposed model. POS can capture syntax feature and first-person pro By incorporating the features, SWNN model can capture both semantic and syntax features.

POS. In Li's analysis [7] between spam reviews and truth reviews, the observations of the POS distribution are in agreement with the early findings in the literature [51,52]. The findings are that truth reviews contain more nouns (N), adjectives (JJ), prepositions (IN) and determiners (DT); spam reviews contain more verbs (V), adverbs (RB), pronouns (PRP) and pre-determiners (PDT). Thus, POS features are meaningful features in distinguishing spam reviews.

First-Person Pronouns. Psychologically, the frequency of the first-person pronouns in a review is relevant to whether the review is a spam or not [52–55]. In the literatures, they find spam reviews contain less first-person pronouns.

3.4. Complexity

We try to make a theoretical analysis in the time complexity of the proposed methods. During learning neural network based models, the time is mainly costed on updating each weight matrix W. Supposed the dimension of W is d, the order of magnitudes is d^2 in the time complexity through each matrix updating. We adopt stochastic gradient descent (SGD) to optimize the parameters which conclude weight matrices. In other words, the matrices are updated at each time when there is a new sample. Supposed the number of samples in dataset is n, then the time complexity of the proposed methods is $\mathcal{O}(n * d^2)$.

4. Experiments

We conduct experiments to empirically evaluate our document representation learning model by applying it in spam review detection. We do two types of experiments which are cross-domain classification and mixed-domain classification. We make an analysis between neural network based methods and SVM with feature combination.

4.1. Data and evaluation criteria

We apply the public dataset released by Jiwei Li [7] which is gold standard spam review dataset. The dataset contains three domains (hotel, restaurant and doctor). The distribution of the dataset is shown in Table 1. There are three types of data in each domain, "Turker", "Expert", and "Customer". They stand for different data sources. The spam reviews are edited by Turkers and experts. Specifically, Li [7] and Ott [5,27] use Amazon Mechanical Turk to collect deceptive reviews from online workers (Turkers). Experts are employees in each domain who have expert-level domain knowledge. The truth reviews are from customers who really have consumption experience.

Table 1
Statistics of the three domain dataset.

Domain	Turker	Expert	Customer
Hotel	800	280	800
Restaurant	200	0	200
Doctor	356	0	200

In the cross-domain classification, we want to make a comparison with Li's method. According to Li's paper, he applies only 200 spam reviews from 356 spam reviews in Doctor domain, and does not apply "Expert" data in his experiment. Hence, we do our best to use data with the same distribution in the cross-domain experiment. We insist the same treatment on positive sentiment samples and negative sentiment samples. Thus, the statistical data in the table is the total number of each domain.

In the mixed-domain spam review classification, all the spam review samples from Turkers and experts in Table 1 and truth reviews from customers are utilized. Then spam reviews are 1636 and truth reviews are 1200. We use five-fold cross validation. The data is split into five equal folds, and four folds are treated as training data, the remaining fold is as test data.

We use accuracy (A), precision (P), recall (R) and F1 score to evaluate the effectiveness of the methods. Accuracy score reflects the prediction capability on both spam samples and non-spam samples. Precision score reflects the correctness of predicting spam samples. Recall score reflects the coverage of correctly predicting spam samples in the true spam samples. F1 score reflects a trade off prediction capability.

4.2. Cross-domain classification

To frame the problem as a domain adaptation task, we want to find a more robust feature on cross-domain dataset. On the latest public data, only Li shows the experiment results. Hence, we do the comparison with his method. Our methods conclude

Table 2
Classifier performance on cross-domain test data.

	Restaurant				Doctor			
Features	A	P	R	F1	A	P	R	F1
Unigram	**0.785**	**0.813**	0.742	**0.778**	0.550	0.573	0.725	0.617
POS	0.735	0.697	0.815	0.751	0.540	0.521	**0.975**	0.679
Paragraph-average	0.733	0.684	0.865	0.764	0.588	0.555	0.885	0.682
Basic CNN+POS+I	0.725	0.679	0.855	0.757	0.583	0.548	0.950	**0.695**
SWNN	0.690	0.644	0.850	0.733	0.610	0.573	0.860	0.688
SWNN+POS+I	0.668	0.612	**0.915**	0.733	**0.615**	**0.576**	0.870	0.693

Table 3
Spam review classification on mixed-domain data.

Model	A	P	R	F1
Unigram	0.825	0.828	0.880	0.853
Bigram+	0.804	0.778	**0.925**	0.845
POS	0.637	0.658	0.776	0.712
POS + bigram+	0.808	0.783	0.924	0.848
SWNN	0.801	0.800	0.873	0.834
SWNN+POS	0.797	0.791	0.886	0.835
SWNN+POS+I	0.822	**0.844**	0.847	0.845
Unigram+I	0.831	0.830	0.890	0.859
Unigram+POS	0.830	0.830	0.887	0.858
Unigram+POS+I	**0.835**	0.839	0.885	**0.861**

paragraph-average, basic CNN, SWNN and the combination of aforementioned methods and features. Specifically, paragraph-average is the average vector of all word embedding in the paragraph which can be regarded as features in the SVM classifier.

Baseline Method. Li respectively apply Unigram, LIWC and POS features in SVM and SAGE classifiers to explore a more general classifier of the task. SAGE is sparse additive generative model which can be viewed as a combination of topic models and generalized additive models. However, SAGE do not outperform SVM. We apply SVM [1] as the classifier in the comparison experiment. In Li's experiment, the method gains best results by using Unigram an POS features in test datasets (restaurant and doctor domains) by training hotel domain data. Hence, we just list the best results from his paper.

Results and Analysis. Table 2 shows the results from baseline method as well as our methods. Unigram get the best result on restaurant domain, but it is not robust on doctor domain. Paragraph-average method performs comparable with the best result on restaurant domain, and is robust across domains. Basic CNN with features perform best on doctor domain. Generally, the neural network based methods perform more robust across domains. We can see SWNN with features does not perform as well as basic CNN with features. The reason is the sentence weight from SWNN is domain-specific.

4.3. Mixed-domain classification

We gather all domain data into a mixed-domain dataset. We verify the effectiveness of proposed neural network method as well as SVM with feature combination. Some comparison experiments are also made among different neural network methods. The experiment results are based on five-fold cross validation.

We adopt previous methods as baseline [5,7], which utilize SVM as classifier with unigram, bigram and POS as traditional features.

Results and Analysis. The results are shown in Table 3, in which unigram with POS and first-person feature gets the best results in accuracy and F1 values. In the spam review detection task, unigram is a strong feature. Even being used alone, unigram has a higher F1 value than SWNN. However, SWNN with features gain highest value in precision which is useful in the application of spam review detection.

[1] We use LIBSVM as the software tool to run SVM classifier.

Table 4
Performance of neural network based method on mixed-domain data.

Model	A	P	R	F1
Paragraph-average	0.729	0.704	0.915	0.795
Weight-average	0.680	0.652	**0.955**	0.775
Basic LSTM	0.550	0.590	0.720	0.720
Hier-LSTM	0.618	0.608	0.949	0.741
Basic CNN	0.708	0.694	0.883	0.776
SCNN	0.702	0.698	0.851	0.766
SWNN	**0.801**	**0.800**	0.873	**0.834**

Table 5
SWNN performance in each domain in spam review classification.

Domain	P	R	F1
Hotel	0.841	0.833	0.837
Restaurant	0.870	0.882	0.876
Doctor	0.850	0.810	0.829

4.3.1. Comparisons among neural network based methods

We apply various neural network based methods to learn the document representation and do the spam review classification. The experiments are gained on mixed-domain dataset by five-fold cross validation.

The Weight-average method is computing a weighted average vector of word embeddings in the document which serves as features in SVM classifier. The weight of each word is computed by information gain. Basic LSTM method uses LSTM to represent sentences and average operation to represent documents. Hier-LSTM use hierarchical LSTM to represent the review documents. Specifically, it utilizes LSTM to composite the sentence representations into document representation. Basic CNN is the basic convolutional neural network model. The sentences are represented through convolutional layer and transform into a document vector by average-pooling operation. SCNN applies convolutional layer to replace the average operation. SWNN is the modification of Basic CNN model by using sentence weights.

Results and Analysis. We do the comparison among various document representations. Table 4 shows the results that our SWNN model gains the best result in deceptive spam classification. The scores of accuracy and F1 are both high above the other neural-network based methods. The results show the effectiveness of incorporating sentence weight in representing document. We also find more complex model like SCNN and Hier-LSTM do not perform as well as simple model like Paragraph-average model and Basic CNN model. Overfitting is a primary reason. Hier-LSTM gains F1 value of 97% on training data, but with a low result on test data. For a small dataset, neural network based models with many parameters is not necessarily a good choice.

Meanwhile, we make an analysis about spam review detection capacity of SWNN on each domain. From Table 5, we find the proposed method performs best on restaurant domain and worst on doctor domain. The reviews on hotel and restaurant domains

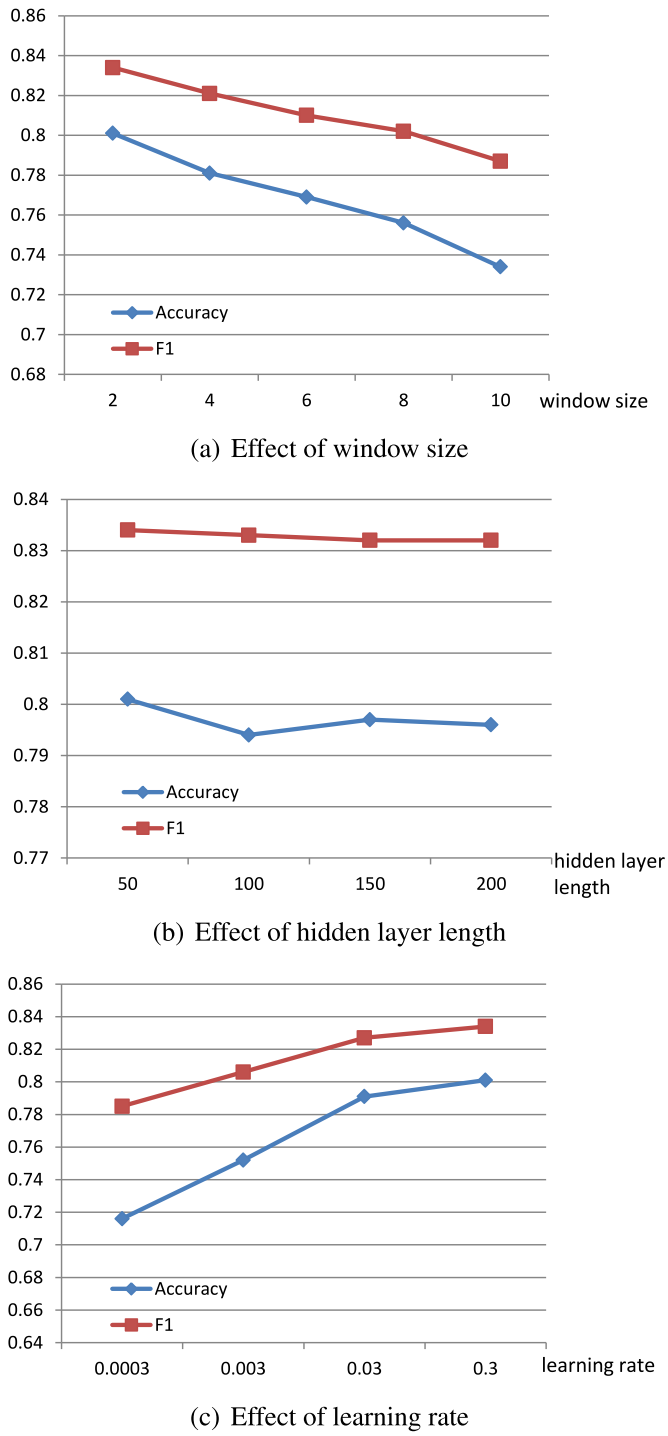

Fig. 4. The effect of three parameters in the experiment.

share more linguistic phenomena [7]. Hence, the model generalizes better on restaurant reviews than on doctor reviews.

4.3.2. Parameter settings

We experimentally study the effect of three parameters in our deceptive spam review detection experiment, which are window size, hidden layer length and learning rate. We do the comparison experiments by five-fold cross validation. The results of averaged accuracy and F1 value are shown in Fig. 4, from which we can see the averaged accuracy and F1 value both have one top when window size is set as 2, hidden layer length is 50 and the learning rate is 0.3. Thus, we use these settings in our experiments.

Table 6
The classification results of SVM with different kernel functions and suitable parameters.

Kernel function	A	P	R	F1	Parameter setting
Linear	0.827	0.838	0.870	0.853	c=1
Polynomial	0.828	0.826	**0.891**	0.857	c=1, g=0.5, r=10, d=3
Gaussian	**0.835**	**0.839**	0.885	**0.861**	c=400, g=0.5
Sigmoid	0.832	0.836	0.883	0.859	c=400, g=0.5, r=0

We use LIBSVM as a software tool to implement SVM classifier. Although there are some optimization approaches [56,57], about the machine learning methods, we tune parameters according to each kernel function by equal distance adjusting method. There are four kernel functions to be chosen in the LIBSVM, which are linear kernel function, polynomial kernel function, gaussian kernel function (also called radial basis function) and sigmoid kernel function. Different kernel functions are used with different parameters. We tune parameter c with linear kernel function; d, g, r and c with polynomial kernel function; g and c with gaussian kernel function; g, r and c with sigmoid kernel function. We tune each parameter for each kernel function. We find that only a part of parameters can affect the classification results based on corresponding kernel function. Specifically, fine tuning of c can enhance the detecting capacity of the models with linear, gaussian or sigmoid kernel function; g and r can enhance the models with polynomial kernel function. The effect of the above parameters in the model with each kernel function is tested by five-fold cross validation. The specific results are shown in the appendix. When we adopt gaussian kernel function and set c as 400, the model has the best results. The results of each kernel function with the most suitable parameters are shown in Table 6.

Finally, we adopt gaussian as the kernel function and set c as 400, g as 0.5, which are verified most effective in the experiments.

5. Conclusion

We introduce a novel convolutional neural network to learn document representation for deceptive spam review detection. Sentences play different roles in the document, in other words they have different importance. We model semantic representation of reviews by incorporating sentence weights into document-level representation learning. We construct experiments on the latest public data sets and compare with multiple baseline methods. The results show that sentence-weighted neural network is more effective than other neural network based models in the deceptive spam review detection. We also find that neural network based methods perform more robust than the hand-crafted features on cross-domain data set. Additionally, we do the comparison between neural network based method and traditional features. By making a feature combination, we enhance the F1 value to 86.1%.

Our sentence-weighted neural network has an attention mechanism which takes the different importance of sentences in the review document into consideration. However, the attention weights are computed by a hard alignment and a fixed mode, which can be improved by a soft alignment and a more flexible mode. The document is intuitively useful through computing the importance weights of each sentence. Thus, memory network based model may be effective to resolve the problem. We will verify the ideas in the future.

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Appendix A. The effect of parameters in SVM

We tune c of SVM separately with linear kernel function, gaussian kernel function and sigmoid kernel function. The effect of c based on each kernel function on spam detection classification is show in Figs. A.1, A.2 and A.3.

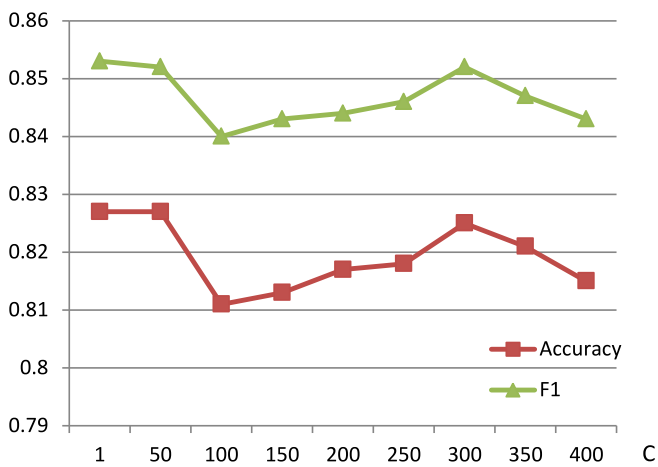

Fig. A1. The effect of c in SVM with linear kernel function.

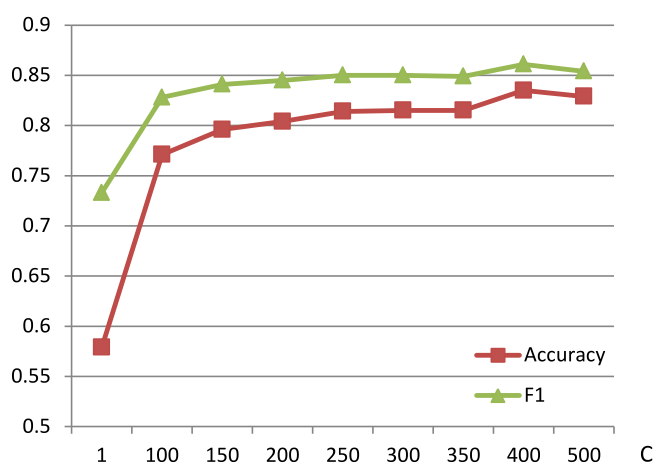

Fig. A2. The effect of c in SVM with gaussian kernel function.

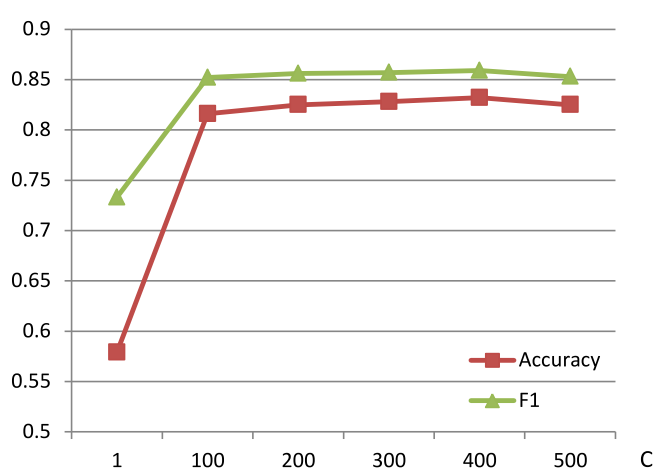

Fig. A3. The effect of c in SVM with sigmoid kernel function.

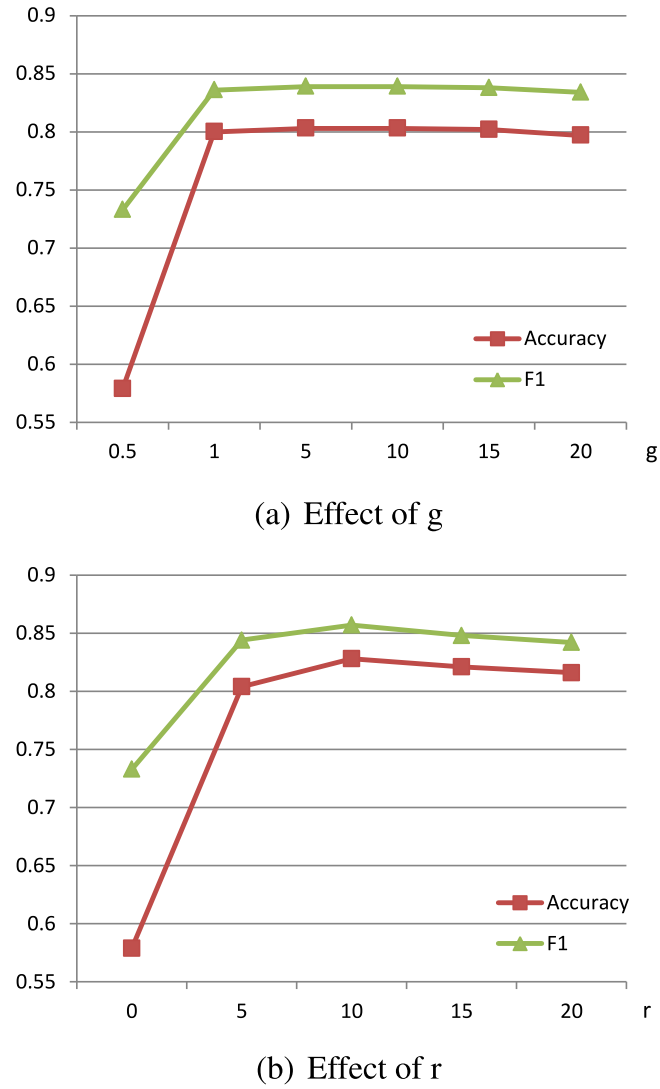

Fig. A4. The effect of g and r in SVM with polynomial kernel function.

We tune d, g, r and c with polynomial kernel function. We find only g and r can affect the results; d and c have no influence. When we tune one parameter, the other parameters are set as default. We also test some combination of the values of g and r. The results show that SVM with polynomial function will acquire good results when r is set as 10 and other parameters are set as default. The effect of g and r on spam detection classification is show in Fig. A.4.

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